Where the cubic-bound proof is right now 2026-06-14 11:20

Working fragment: classical regexes, in normal form, no bounded repetition $r^{\{n\}}$ (the "clean" fragment). Everything below is machine-checked in Isabelle with no gaps unless marked [$\to$] or [$\circ$].

The goal (one inequality). Run the lexer's normalized derivative automaton from r over input s, do *one* strongly-simplified step on the next letter c, open every resulting row into linear forms and deduplicate, and sum their sizes. That total is cubic in $|r|$:

$$\boxed{\|\ \mathrm{Open}\big(\Delta_c(\mathrm{Front}(r,s))\big)\ \|\ \le\ 2\,(|r|+3)^3}$$

uniformly in s (its length does not appear on the right).

Notation. $|r|$ = size of r; $\mathrm{acc}(r,k)$ = rows the derivative of r leaves in front of continuation k; ∂k = one-step frontier of k; σ = sequence normalizer; $\mathrm{w}(r)$ = weight (letters, +1 per star) — always $\mathrm{w}(r) \le |r|$; $\mathrm{Rows}(r)$ = all rows reachable from r (static); $\mathrm{Front}(r,s)$ = the normalized front after reading s; Δ_c = one strong simplify+prune step; Open = open rows to linear forms and dedup.

The chain (top to bottom). Read it as: each line is justified by the one below it.

[✓ PROVEN] **Goal, on the clean fragment.** $\|\mathrm{Open}(\Delta_c(\mathrm{Front}(r,s)))\|$ splits into a non-sequence part (already cubic) plus a sequence part bounded by the front total $\|\mathrm{Front}(r,s)\|$. actual_union_gate_two_summands; rsize_set_rnonseq_members_. . . _cubic

[✓ PROVEN] **Static front is cubic, for every input.**

$$\|\mathrm{Front}(r,s)\|\ \le\ (|r|+3)^3 \qquad \text{for all } s.$$

In words: however long the input, the normalized front never exceeds a fixed cubic in the original size. rsizes_afactored1_clean_static_cubic

[✓ PROVEN] **It rests on two facts about the static row universe:**

$$\#\,\mathrm{Rows}(r)\ \le\ |r|+2 \qquad \text{and} \qquad \text{every row in } \mathrm{Rows}(r) \text{ has size } \le\ (|r|+2)^2.$$

In words: few rows (linearly many), each small (quadratic). Linear × quadratic = cubic. card_apder_rows_clean_le_rsize_plus_2; apder_rows_member_size_quadratic

[✓ PROVEN] **The row count is linear because of the D law** (the hard part, cracked via the GPT-Pro "telescoping" invariant):

$$\#\big(\mathrm{acc}(r,k)\setminus\partial k\big)\ \le\ \mathrm{w}(r) \qquad (\text{clean } r,k).$$

In words: one derivative step adds at most $\mathrm{w}(r)$ ($\le |r|$) genuinely new rows. Proved by a single simultaneous induction on the stronger boundary invariant $\#((\partial(\sigma r k)\cup \mathrm{acc}(r,k))\setminus\partial k) \le \mathrm{w}(r)$ plus a strict-credit lemma. D_law_clean (from apder_T_bound/apder_S_bound via T_and_S)

[✓ PROVEN] **The D law applies to the *actual* rows.** The rows the gate feeds are in the clean domain. apder_clean_apder_rows; apder_clean_afactored1

[$\to$ CURRENT] **The one remaining step: glue the cubic front into the gate.** Push $\|\mathrm{Front}(r,s)\| \le (|r|+3)^3$ through the single strong step, the opening, and the dedup

$$\mathrm{Front}(r,s)\ \xrightarrow{\Delta_c}\ \cdot\ \xrightarrow{\mathrm{Open}}\ \cdot\ \xrightarrow{\|\cdot\|}\ \le\ 2(|r|+3)^3,$$

and discharge the "clean" hypothesis for the actual root. **Design point, status 2026-06-15.** The D law and the static cubic front are done; the §5 drain arithmetic is green; all six per-constructor cover cases and the gate WRAPPER are green. The §1 gate reduces (proven wrapper) to ONE numeric bound: rsize_set(strong_child_drain p k) $\leq$ ctx_bound(drain_ctxs p) k, which is machine-validated TRUE (zero violations over $> 10^5$ depth-$\geq$5 cases). *But the PROOF is the stubborn part*: FIVE structural routes have now been depth-$\geq$5 REFUTED, all on the same guarded collapse rsimp7_SEQ_atom$(a^*, a^*) = a^*$ — per-child set-containment (the cases do not compose), the recursive scover, the injective slot-origin charge via collapses_to and a two-route variant, and a pure numeric induction (the RALTS step fails). *The sharp diagnosis:* strong_child_drain mixes S-collapsed short rows and uncollapsed long rows (RALTS re-exposure), and NO injective row→slot charge can exist — the total ledger budget covers the rows, but some slots are overloaded while others are underused (a Hall collision with a true size bound). The remaining design problem (a parallel candidate search + a GPT Pro pass): a **non-injective / aggregate / total-budget** proof of the (true) bound. Once it lands, strong_child_drain_potential + the gate wrapper close the boxed goal (clean fragment).

[○ LATER] **After that:** lift the result from the clean fragment to *all* regexes including bounded repetition $r^{\{n\}}$. This needs a repetition-aware weight (the plain w fails when a $\{0\}$ block carries an empty-but-present frontier). Optional; may not be required for the thesis target.

The symbols, defined (so the inequalities above are self-contained). Throughout, r, k are regexes, Σrs a list-alternation, c a letter.

- **Size** $|r|$: $|0| = |1| = |c| = 1$, $|r_1 r_2| = 1 + |r_1| + |r_2|$, $|\Sigma rs| = 1 + \sum_i |r_i|$, $|r^*| = 1 + |r|$, $|r^{\{n\}}| = 1 + |r| + n$.
- **Frontier** ∂k — the rows of k after splitting its top alternation: $\partial 0 = \emptyset$, $\partial(\Sigma rs) = \bigcup_i \partial r_i$, $\partial k = \{k\}$ otherwise. *(A plain row is itself; an alternation splits into its branches.)*
- **Sequence normalizer** $\sigma(r, k)$ — "the row r followed by k," with units erased and right-associated: $\sigma(0, k) = 0$, $\sigma(1, k) = k$, $\sigma(r_1 r_2, k) = \sigma(r_1, \sigma(r_2, k))$; for an atom a (letter / star / alternation): $\sigma(a, 0) = 0$, $\sigma(a, 1) = a$, else $\sigma(a, k) = a\,k$.
- **Accumulator** $\mathrm{acc}(r, k)$ — the rows that one derivative of r leaves in front of continuation k:

$$\mathrm{acc}(0, k) = \mathrm{acc}(1, k) = \emptyset, \quad \mathrm{acc}(c, k) = \partial k, \quad \mathrm{acc}(\Sigma rs, k) = \bigcup_i \mathrm{acc}(r_i, k),$$

$$\mathrm{acc}(r_1 r_2,\, k) = \mathrm{acc}\big(r_1, \sigma(r_2, k)\big) \cup \mathrm{acc}(r_2, k), \qquad \mathrm{acc}(r^*, k) = \mathrm{acc}\big(r, \sigma(r^*, k)\big).$$

  (A letter hands its whole continuation-frontier forward; a sequence accumulates the left part against the normalized $r_2 k$ plus the right part against k; a star re-enters itself. This recursion — one letter importing a whole frontier, balanced only across siblings — is exactly what made the D law hard.)

- **Weight** $\mathrm{w}(r)$ — letters, +1 per star; always $\mathrm{w}(r) \leq |r|$: $\mathrm{w}(0) = \mathrm{w}(1) = 0$, $\mathrm{w}(c) = 1$, $\mathrm{w}(r_1 r_2) = \mathrm{w}(\Sigma rs) = \text{sum}$, $\mathrm{w}(r^*) = 1 + \mathrm{w}(r)$, $\mathrm{w}(r^{\{n\}}) = n\,(1 + \mathrm{w}(r))$.
- **Static row universe** $\mathrm{Rows}(r) = \{r\} \cup \partial r \cup \bigcup_q \partial q$, the union over all Antimirov sub-derivative terms q of r. *(Every row that can ever appear in front of r; a finite set.)*

One-line status: **the hard combinatorics (the D law) and the static cubic front are done; one mechanical gluing step + a root-hypothesis discharge remain to close the gate on the clean fragment.** Full formal statement: `CUBIC_OPEN_PROBLEM.pdf`. Live detail: `MAINLINE.md` §2 and the `PROGRESS_BACKREF.md` tail.